

Transformations of Random Variables

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Introduction

Given a random variable X and a function g , the distribution of $Y = g(X)$ is determined by the distribution of X and the function. The change-of-variables formula computes the new distribution; for strictly monotone differentiable g this involves a Jacobian factor.

Prerequisites

Random variables, PDF, basic calculus.

Theory

Univariate, monotone case. Let X have density f_X on support $\mathcal{X}$, and let g be strictly monotone and differentiable on $\mathcal{X}$. Then $Y = g(X)$ has density

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|.$$

The absolute value of the derivative of the inverse is the **Jacobian**.

Non-monotone case. If g is piecewise monotone, partition the domain into monotone segments and sum the contributions:

$$f_Y(y) = \sum_i f_X(x_i) / |g'(x_i)|$$

where $\{x_i\}$ are the solutions of $g(x) = y$.

Multivariate case. For $\mathbf{Y} = g(\mathbf{X})$ with $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ a diffeomorphism,

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(g^{-1}(\mathbf{y})) |\det J_{g^{-1}}(\mathbf{y})|,$$

where $J_{g^{-1}}$ is the Jacobian matrix of g^{-1} .

Common examples:

- $Y = aX + b$: $f_Y(y) = f_X((y - b)/a) / |a|$.
- $Y = X^2$, X standard normal: $Y \sim \chi_1^2$.
- $Y = e^X$, X normal: Y is log-normal.
- $Y = F^{-1}(U)$, $U \sim \text{Uniform}(0, 1)$: the inverse-CDF method for generating arbitrary distributions.

Assumptions

- g must be measurable.
- For the change-of-variables formula to apply, g must be strictly monotone and differentiable (or piecewise so).

R Implementation

```
set.seed(2026)

# Case 1:  $Y = X^2$  for  $X \sim N(0, 1)$ ;  $Y$  should be chi-squared with 1 df
X <- rnorm(1e5)
Y <- X^2

par(mfrow = c(1, 2))
hist(Y, breaks = 100, freq = FALSE, main = "Empirical", col = "#2A9D8F")
curve(dchisq(x, df = 1), from = 0, to = 10, add = TRUE, col = "#F4A261", lwd = 2)

# Case 2:  $Y = \exp(X)$  for  $X \sim N(\mu, \sigma)$  is log-normal
mu <- 1; sigma <- 0.5
X2 <- rnorm(1e5, mu, sigma)
Y2 <- exp(X2)
curve(dlnorm(x, mu, sigma), from = 0, to = 10,
      main = "exp(N(1, 0.25)) vs lognormal PDF",
      col = "#2A9D8F", lwd = 2)
hist(Y2, breaks = 100, freq = FALSE, add = TRUE, border = "#F4A261")

par(mfrow = c(1, 1))

# Case 3: Inverse-CDF method to generate exponential from uniform
U <- runif(1e5)
E <- -log(1 - U) #  $F^{-1}(u) = -\log(1 - u)$  for rate = 1
c(mean = mean(E), theoretical = 1)
```

Output & Results

mean	theoretical
0.999	1.000

Empirical histograms match the theoretical transformed densities. The inverse-CDF method recovers the exponential mean.

Interpretation

Many “derived” distributions in statistics (chi-squared from normals, log-normal from normals, Cauchy from normals, Beta from gammas) arise as transformations of simpler distributions. Recognising the transformation short-cuts many derivations.

Practical Tips

- Always check strict monotonicity before applying the simple Jacobian formula; square-mapping X^2 is a classic non-monotone case.
- For inverse-CDF simulation, you need the CDF’s inverse; use `qnorm`, `qgamma`, etc., in R.
- The Jacobian for a multivariate linear transformation $Y = AX$ is $|\det A|$, independent of location.
- Delta method is the “linearised” version of the change-of-variables formula, used for asymptotic distributions.
- Non-bijective transformations (e.g., $|X|$) require summing over preimages, as in the non-monotone case.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots